JSS MAHAVIDHYAPEETHA
JSS SCIENCE AND TECHNOLOGY UNIVERSITY
SRI JAYACHAMARAJENDRA COLLEGE OF ENGINEERING

- Constituent College of JSS Science and Technology University
- Approved by A.I.C.T.E
- Governed by the Grant-in-Aid Rules of Government of Karnataka
- Identified as lead Institution for World Bank Assistance under TEQIP Scheme

DEPARTMENT OF ELECTRICAL AND ELECTRONICS ENGINEERING

SCHEME & SYLLABUS

First Semester

(From the Academic Year 2022-23)

Course Title	**Elements of Electrical Engineering**			**Course Type**		**Theory**			
Course Code	**24EE110/ 24EE210**	**Credits**	**3**	**Class**		**I Semester /II Semester**			
Course Structure	TLP	Credits	Contact Hours	Total Number of Classes / Semester		Assessment inWeightage and Marks			
	Theory	3	3 h/w	**Theory**	**Practical**	**CIE**		**SEE**	
	Practical	0	0	-	-	**Weightage**	40%	**Weightage**	60%
	Tutorial	0	0	-	-	**Max. Marks**	40 Marks	**Max. Marks**	60 Marks
	Total	**3**	**39**	**39**	**-**	**Min. Marks**	20 Marks	**Min. Marks**	25 Marks

Course overview: This course covers the fundamental concepts of electrical circuits, single phase transformers, alternators, and induction motors. The course also introduces the fundamental aspects of electrical installations, safety and electrical power systems

Pre requisites: Nil

Course Objective: To make the students get familiarized with the fundamentals of electric circuits, working principles of electric machines, electric power generation and domestic electrical installations. Also to enable students to solve simple problems on electric circuits and machines.

Course Outcome: After completing this course, the student should be able to:

CO#	Course Outcome	Highest Level of Cognitive Domain
CO1	Explain the concepts of power generation and grid interconnection in an electrical power system	**L2**
CO2	Describe the aspects of domestic electrical installation and safety.	**L2**
CO3	Apply the fundamental concepts to solve DC and Electromagnetic circuits.	**L3**
CO4	Apply the fundamental electrical concepts to solve AC circuits.	**L3**
CO5	Explain the construction, types, working principle of Alternators, Transformers and Induction motors.	**L2**

L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate, L6 - Create

Course Articulation:

POs → COs ↓	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	-	-	-	-	-	1	-	-	-	-	-
CO2	3	-	-	-	-	-	-	-	-	-	-	1
CO3	3	3	-	-	-	-	-	-	-	-	-	-
CO4	3	3	-	-	-	-	-	-	-	-	-	-
CO5	3	3	-	-	-	-	-	-	-	-	-	-

High – 3, Medium – 2, Low – 1

Course Content / Syllabus:

UNIT No.	Content	Hours	
UNIT 1	**INTRODUCTION TO ELECTRICAL POWER SYSTEMS** **Generation:** Conventional and Non-conventional energy sources, generation of power from Hydro, thermal (coal based), nuclear, solar and wind (Block diagram approach), Distributed generation and its benefits. **Transmission and Distribution:** Primary & Secondary transmission and distribution voltage levels with Single line diagram, concept of grid and the need for interconnection of grids. Conditions for grid Connection, Smart grid- Definition and benefits. **Regulatory bodies-** Central and State in electrical power sector and their roles. Power tariff- Definition and types.	Lecture 7	Tutorial -
UNIT 2	**ELECTRICAL INSTALLATIONS** Typical domestic wiring circuit, Underground cables- single core and four core only, Types of lamps and its applications (Fluorescent, Compact Fluorescent and LED). Types of switches, two position and three position control of switches, Electric Shock and its prevention. Need for Earthing and grounding, Procedure of Pipe earthing. Protective devices-definition, rating and working (Fuse, MCB and ELCB). UPS – working of Online and Offline systems (Block diagram approach), Working of Electronic and Smart Energy meters (Block diagram approach).	Lecture 7	Tutorial -

UNIT 3	**DC Circuits and Electromagnetism** **D.C. Circuits:** Ohm's Law, Kirchhoff's Laws and their application to solve problems. **Electromagnetism**: Faraday's laws, Lenz’s law, Fleming’s rules, Concept of self-inductance, mutual-inductance and mutual coupling, Illustrative examples.	Lecture 7	Tutorial -
UNIT 4	**AC Circuits** **Single Phase circuits:** Generation of alternating quantity and its terminologies, phasor representation. Analysis of series circuits(R, L, C, R-L, R-C, R-L-C). Analysis of power triangle and power factor, Illustrative Examples. **Three Phase Circuits:** Necessity and advantages of three phase systems, generation of three phase power. Definitions of Phase sequence, balanced supply and balanced load. Relationship between voltage, current and power for balanced star and delta connections (Excluding derivations). Illustrative examples.	Lecture 8	Tutorial -
UNIT 5	**Electrical Machines** **Alternators:** Construction, types, and Principle of operation. **Transformers:** Construction and principle of operation of single-phase transformers, types, EMF equation, losses, efficiency, ratings/specifications, applications, standard codes, Illustrative examples. **Three Phase Induction Motors:** Construction and principle of operation, types, slip-torque characteristics, applications, Necessity of a starter.	Lecture 10	Tutorial -

Text Books:

1. Edward Hughes, Dr. John Hiley, Ian McKenzie-Smith, and Dr. Keith Brown, “ Electrical and Electronic Technology”, 10th Edition, Pearson, 2010.
2. D. P. Kothari and I. J. Nagarath, “Basic Electrical and Electronics Engineering”, 3rd Reprint, 2016.
3. D.C. Kulshreshtha, “Basic Electrical Engineering”, TMH, Revised First Edition.
4. Lecture Material /Course Material

Reference Books:

1. V K Mehta and Rohit Mehta, “Basic Electrical Engineering”, Sixth Edition , S Chandh.
2. N. N. Parker Smith, “Problems in Electrical Engineering”, Ninth Edition, CBS Publishers, New Delhi.

Web/Digital resources:

https://nptel.ac.in/courses/108/105/108105053

Evaluation Scheme:

Continuous Internal Evaluation – CIE

Event	Event Type	Marks Allotted	Duration
CIE – 1	Written Test – 1	30	1 Hour
CIE – 2	Event	20	1 Hour
CIE – 3	Written Test – 2	30	1 Hour

Note:

1. The written test 1 & 2 (CIE – 1 & 3) both will be conducted for 30 marks each in 1 hour duration and the marks scored will be reduced proportionately to 15 marks each.
2. The Event (CIE – 2) will be conducted for 20 marks and the marks scored will be reduced proportionately to 10 marks.
3. The Event (CIE – 2) will be skill based assessment such as Seminars / Technical talks / Case study / hands-on activity / Mini projects / Sci-tech activity / Data analysis.
4. A student must score on an average of 50% i.e., 20 marks out of 40 from all the events (CIE - 1, 2, 3) to gain the eligibility to appear for SEE.

Semester End Examination – SEE

Event	Event Type	Marks Allotted	Duration
SEE	Written Examination	100	3 Hours

Note:

1. The SEE will be conducted for 100 marks and the marks scored will be proportionately reduced to 60 marks. A student must score a minimum of 25 marks out of 60 in the SEE to pass.
2. SEE Question paper will be set for 100 marks and will have two parts (Part-A & Part-B). Questions under Part-A are compulsory and questions under Part-B will have internal choices.